工程數學 Midterm #2 Dec, 6 2022

1. 5pt $\mathrm{L}\{(\mathrm{a}-\mathrm{bt})^2\}$, a and b are constant	7. 10pt $y'' + 2y' - 3y = 0,$ $y(2) = -3, \quad y'(2) = -5$
2. 10pt $L^{-1}\left\{\frac{6s+7}{2s^2+4s+10}\right\}$	8. 10pt $L^{-1}\left\{\frac{2}{s^2+\frac{s}{3}}\right\}$
3. 5pt $\mathrm{L}\{10e^{-3t} - 5\mathrm{t} + 8\}$	9. 10pt $y'' + 4y' + 6y = 1 + e^{-t},$ $y(0) = 0, \quad y'(0) = 0$
4. 10pt $L^{-1}\left\{\frac{0.2s + 1.4}{s^2+1.96}\right\}$	10. 10pt $L^{-1}\left\{\frac{5s}{(s-2)^2}\right\}$
5. 10pt $L^{-1}\left\{\frac{11-s}{s^2-2s-3}\right\}$	11. 10pt $L^{-1}\frac{1}{s^3+s}$ $x^2y'' = D_t(D_t-1)y$ $x^3y''' = D_t(D_t-1)(D_t-2)y$
6. 10pt $L^{-1}\left\{\frac{s}{s^2+4s+5}\right\}$	

- Laplace 轉換

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$	7	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
2	t	$1/s^2$	8	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
3	t^2	$2!/s^3$	9	$\cosh at$	$\frac{s}{s^2 - a^2}$
4	t^n $(n = 0, 1, \cdots)$	$\frac{n!}{s^{n+1}}$	10	$\sinh at$	$\frac{a}{s^2 - a^2}$
5	t^a (a positive)	$\frac{\Gamma(a + 1)}{s^{a+1}}$	11	$e^{at} \cos \omega t$	$\frac{s - a}{(s - a)^2 + \omega^2}$
6	e^{at}	$\frac{1}{s - a}$	12	$e^{at} \sin \omega t$	$\frac{\omega}{(s - a)^2 + \omega^2}$

First Shifting Theorem, s-Shifting $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$ $e^{at}f(t) = \mathcal{L}^{-1}\{F(s - a)\}$	Laplace Transform of Derivatives $\mathcal{L}(f') = s\mathcal{L}(f) - f(0)$ $\mathcal{L}(f'') = s^2\mathcal{L}(f) - sf(0) - f'(0)$ $\mathcal{L}(f^{(n)}) = s^n\mathcal{L}(f) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$

Laplace Transform of Integral 積分轉換

$$\mathcal{L}\left\{\int_0^t f(\tau)\, d\tau\right\} = \frac{1}{s}F(s), \qquad \text{thus} \qquad \int_0^t f(\tau)\, d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}F(s)\right\}$$